The State of Road Safety in the City of Toronto*

Analyzing Vehicle Collisions from 2006 to 2026

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1 Introduction

Home to over 3 million residents, Toronto is the most populated and one of the densest cities in Canada. Boasting a metropolitan center this large, there is constant movement across the city from pedestrians to drivers. However, Toronto faces the challenge of resident safety when it comes to road transportation. While vehicle collisions are an unfortunate reality of urban transportation, municipal data shows road safety as a severe and growing public issue. Even as of late, the heated debates on the inclusion of speeding cameras within the province show how road safety is important to many. To tackle these dangers on the road, the city developed the Vision Zero Road Safety Plan and actively provides public, data-driven dashboards tracking total collision counts. However, while raw data is highly accessible, there is a gap in understanding geographic disparities and the specific profiles of the users most impacted by these collisions.

To address this, we use 20 years of open-source Motor Vehicle Collision data from the City of Toronto Open Data Portal to investigate the locations and timing of roughly 20,000 road incidents. The scope of the data allows us to evaluate the realities of road safety over time.

We find that severe incidents peak sharply during the morning and evening rush hours, showing how risks are tied to standard commuter movement. Furthermore, while the absolute count of collisions varies widely across wards, population-adjusted rates reveal spatial inequities, with specific neighborhoods experiencing higher risks. Overall, where and how people drive plays into the safety of both drivers and local residents, making it important to also curate targeted interventions across specific neighbourhoods while planning overarching safety campaigns.

The remainder of this paper is structured as follows: Section 2 describes the data source. Section 3 presents the results of the geographic and temporal analyses. Section 4 discusses the policy implications of these findings.

*Code and data are available at: [https://github.com/arusansurendiran/toronto-road-safety).

2 Data

2.1 Overview

We use the statistical programming language R (R Core Team 2023)…. Our data (Toronto Shelter & Support Services 2024)…. Following Alexander (2023), we consider…

Overview text

2.2 Measurement

Some paragraphs about how we go from a phenomena in the world to an entry in the dataset.

2.3 Outcome variables

Add graphs, tables and text. Use sub-sub-headings for each outcome variable or update the subheading to be singular.

Some of our data is of penguins (**?@fig-bills**), from Horst, Hill, and Gorman (2020)

3 Results

Our results are summarized in **?@tbl-modelresults**.

Table 1: Types of Collision

Accident Class	Total Count	Proportion
Non-Fatal Injury	17492	0.859
Fatal Injury	2866	0.141

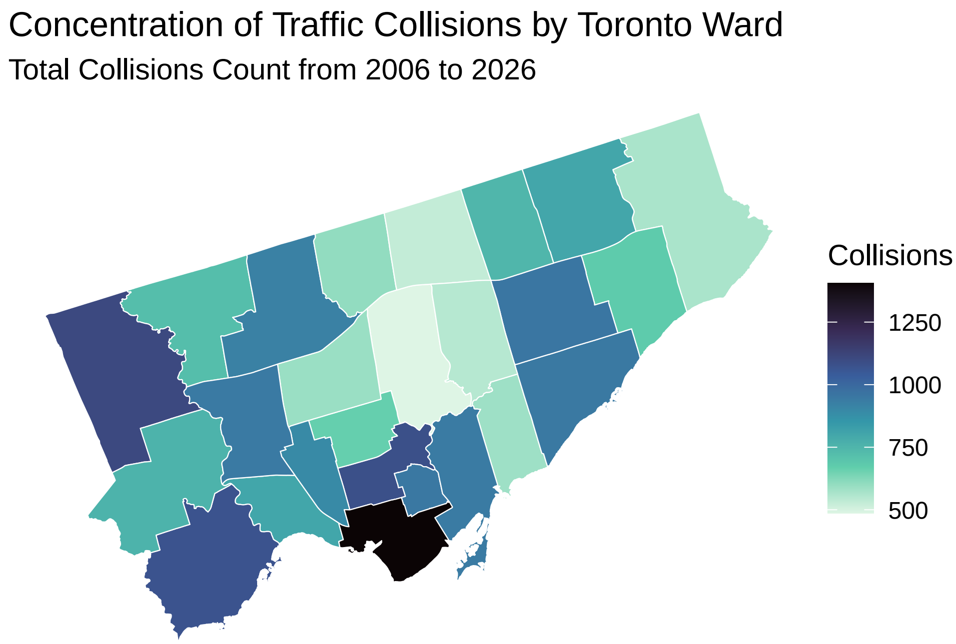

Figure 1: Bills of penguins

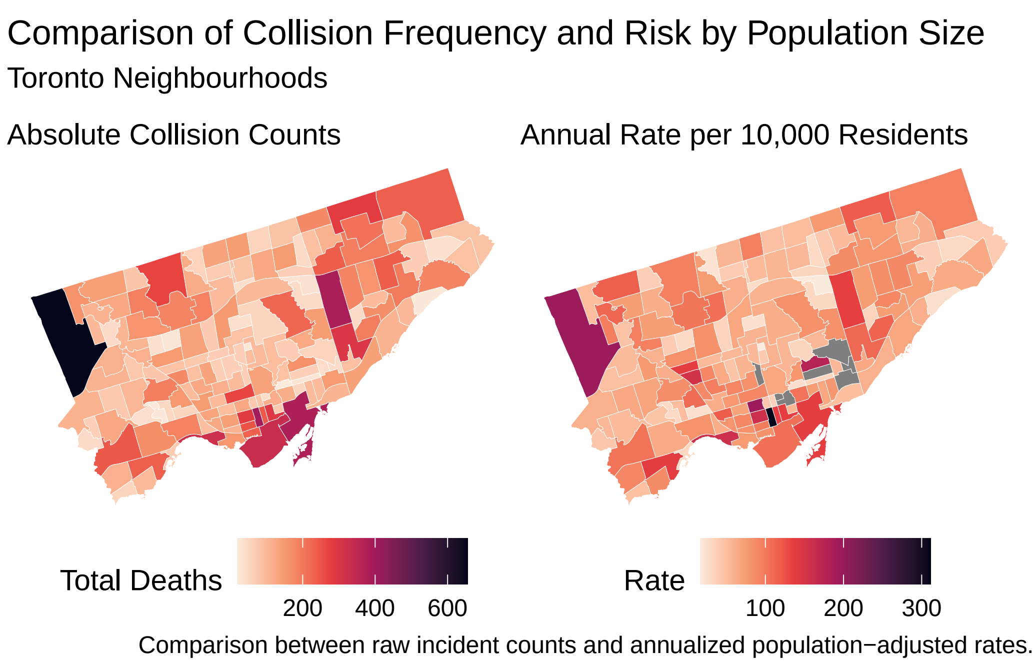

Figure 2: Bills of penguins

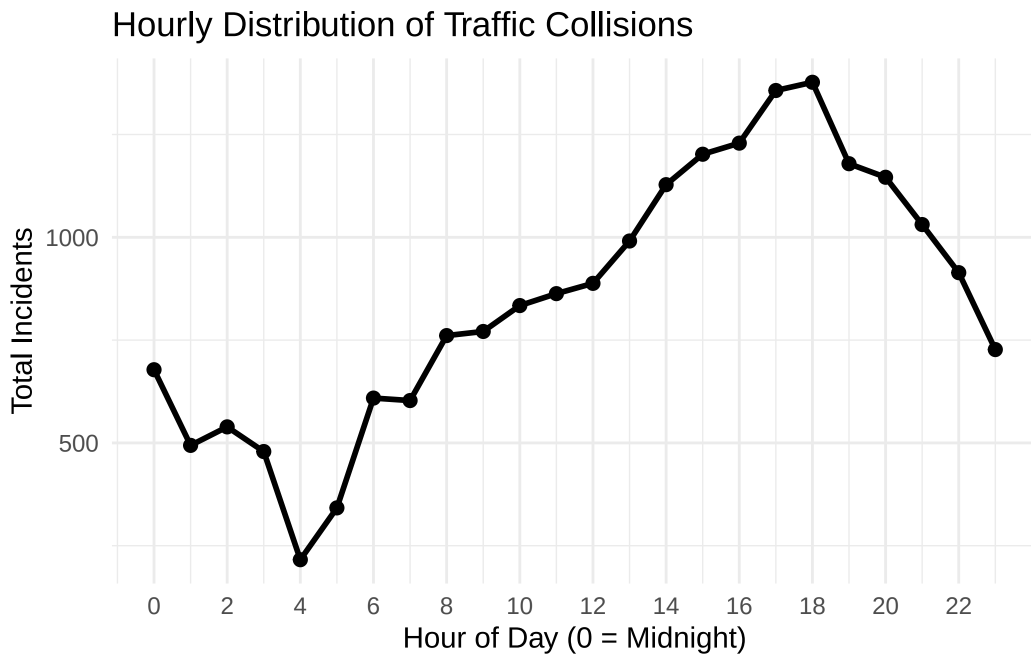

Figure 3: Bills of penguins

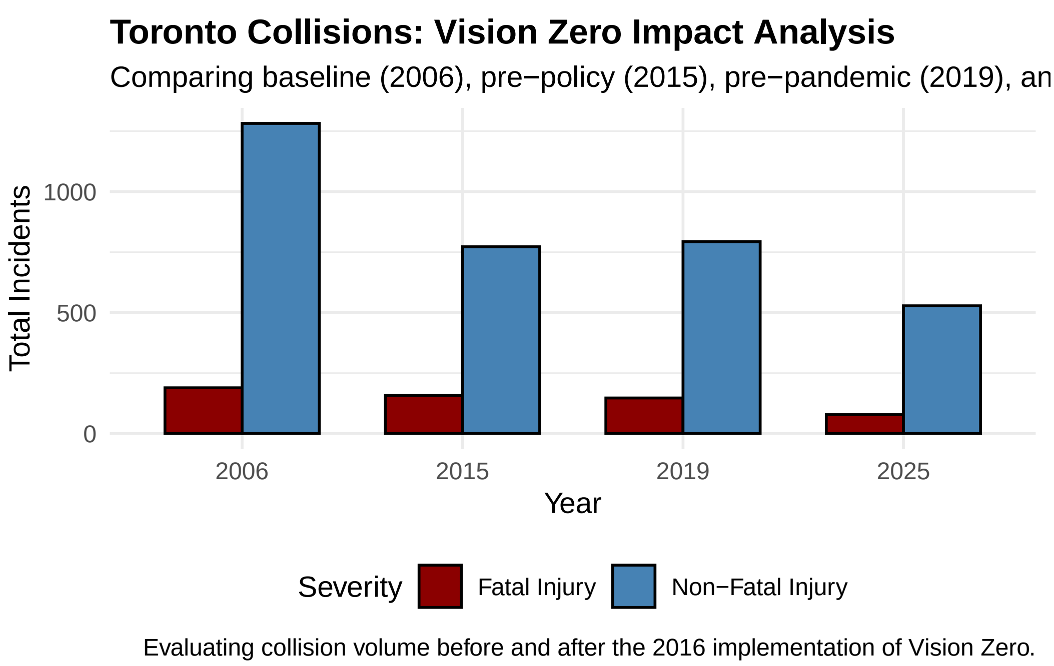

Figure 4: Bills of penguins

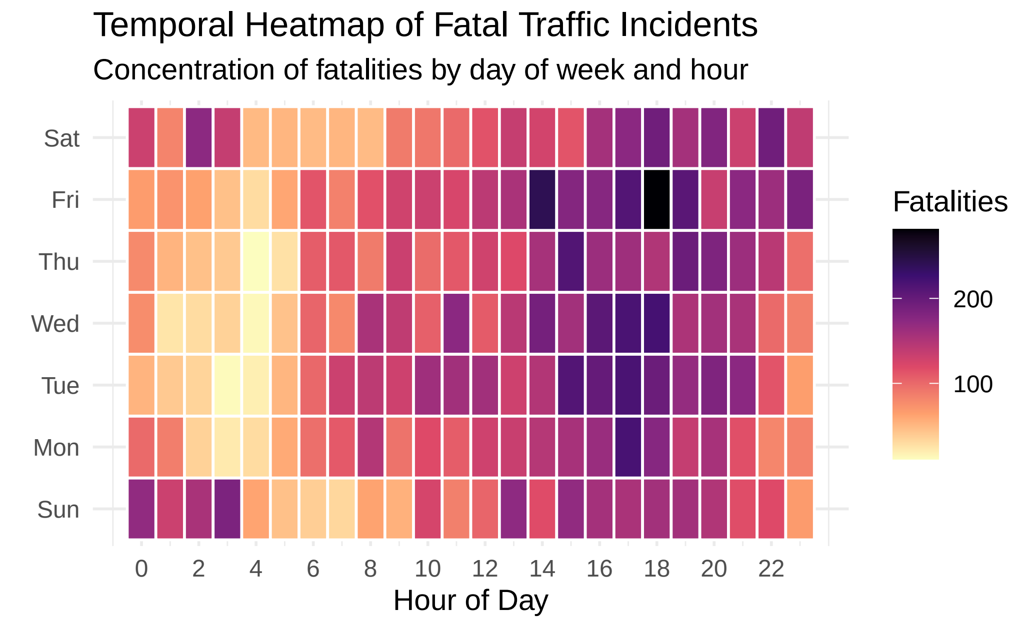

Figure 14: Heatmap visualization identifying critical risk windows.

Figure 5: Bills of penguins

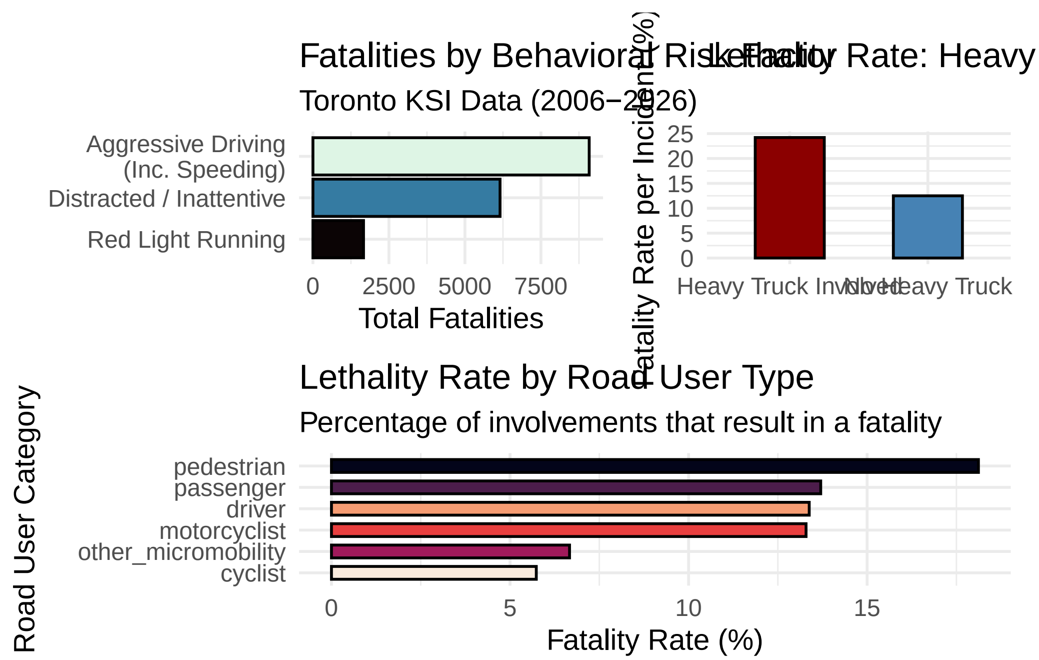

Figure 6: Bills of penguins

Figure 1: Concentration of Traffic Collisions by Toronto Ward

- absolute count of traffic collisions is heavily concentrated in the downtown core
- outer wards generally show lower absolute collision volumes compared to the central downtown hub.

Figure 2: Comparison of Collision Frequency and Risk by Population Size (Neighbourhoods)

- absolute collision counts highlight downtown and specific western areas
- adjusting for population size makes some shifts to the risk map, noting how Scarborough area has more comparable level of incidents compared to the rest of Toronto (still higher though)
- annualized rate per 10,000 residents shows that several outer neighborhoods, like in the northwest, experience high collision rates
- the northwest of toronto is also more typically associated with other deep inequites too

Figure 3: Hourly Distribution of Traffic Collisions

- Collision frequency is really high during peak 4-6 rush house standard commuter traffic
- sustained increase throughout day hours, then dips after 6pm
- incident volumes hit their lowest point at 04:00 before traffic begins to increase

Figure 4: Vision Zero Impact

- reduction in overall collision volume when comparing the 2006 baseline to recent years.
- remained flat between the pre-policy (2015) and pre-pandemic (2019), maybe saying early Vision Zero implementations did not yield good structural changes
- by 2025, both total non-fatal and fatal injuries show a decrease.

Figure 5: Week Day and Hourly Distribution

- you can see that friday and saturday nights, especially in the AM times of those days, shows an increase in collisions
- aligns with the general idea of more freetime for people from weekdays, suggesting later times of going out and possible more reckless behvaiour on roads
- weekdays are relatively quieter in the AM pre-dawn hours

Figure 6:

- aggressive driving is the most common cause

- trucks are in general more dangerous
- pedestrians are the most at risk from vehicle collisions across the city

4 Discussion

Appendix

A Data

A.1 Additional Data Details

A.2 Data Cleaning

References

Alexander, Rohan. 2023. *Telling Stories with Data*. Chapman; Hall/CRC. https://tellingstorieswithdata.com/.

Horst, Allison Marie, Alison Presmanes Hill, and Kristen B Gorman. 2020. *palmerpenguins: Palmer Archipelago (Antarctica) penguin data*. https://doi.org/10.5281/zenodo.3960218.

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